Selected Subjects in Robotics and AI HW1

Daniel Haim Breger, 316136944
Ethan Chelly, 346395064

Technion

(Date: December 4, 2025)

1 Maximum Likelihood Estimation for Non-Gaussian DIstribution

Let $(X_1, X_2, \dots, X_n)$ be i.i.d samples from a Rayleign distribution with unknown scale parameter $(\theta > 0)$:

$$f(x, \theta) = \frac{x}{\theta^2} e^{-x^2/(2\theta^2)}, \quad x \geq 0$$

1. Derive the maximum likelihood estimator $\hat{\theta}_{ML}$.
2. Show that it is a consistent estimator of θ.
3. Determine whether $\hat{\theta}_{ML}$ is biased. If it is, derive the bias expression.
4. Compute the Fisher Information and Cramer-Rao lower bound for estimating θ and check if $\hat{\theta}_{ML}$ attains it asymptotically.

1.1 Deriving MLE

The log-likelihood is

$$l(\theta) = \sum_{i=1}^{n} \left(\log X_i - 2 \log \theta - \frac{X_i^2}{2\theta^2} \right) \tag{1}$$

The definition of the MLE is

$$\hat{\theta}_{MLE} = \arg\max L(\theta, x) \tag{2}$$

To find the argmax we'll differentiate the log-likelihood:

$$\frac{\mathrm{d}l}{\mathrm{d}\theta} = \sum_{i=1}^{n} \left(-\frac{2}{\theta} + \frac{X_i^2}{\theta^3} \right) = \frac{-2n}{\theta} + \frac{1}{\theta^3} \sum_{i=1}^{n} X_i^2 \tag{3}$$

we require the derivative is equal to zero and find

$$\frac{\mathrm{d}l}{\mathrm{d}\theta} = \frac{-2n}{\theta} + \frac{1}{\theta^3}\sum_{i=1}^{n} X_i^2 = 0 \tag{4}$$

$$\rightarrow -2n\theta^2 + \sum_{i=1}^{n} X_i^2 = 0 \tag{5}$$

$$\rightarrow \theta_{\max} = \sqrt{\frac{1}{2n}\sum_{i=1}^{n} X_i^2} \tag{6}$$

verifying the point we found is indeed a maximum

$$\left.\frac{\mathrm{d}^2 l}{\mathrm{d}\theta^2}\right|_{\theta_{\max}} = \left.\frac{2n}{\theta^2} - \frac{3}{\theta^4}\sum_{i=1}^{n} X_i^2\right|_{\theta_{\max}} \tag{7}$$

$$= \frac{2n}{\theta_{\max}^2} - \frac{3}{\theta_{\max}^4}\cdot 2n\theta_{\max}^2 \tag{8}$$

$$= \frac{2n - 6n}{\theta_{\max}^4} = -\frac{4n}{\theta_{\max}^4} < 0 \tag{9}$$

Therefore it is indeed a maximum, so our MLE is

$$\hat{\theta}_{MLE} = \sqrt{\frac{1}{2n}\sum_{i=1}^{n} X_i^2} \tag{10}$$

1.2 Consistency Check

An estimator is called consistent if

$$\lim_{n\to\infty} \hat{\theta}_n = \theta \tag{11}$$

So we'll look at the limit

$$\lim_{n\to\infty} \hat{\theta}_{MLE} = \lim_{n\to\infty} \sqrt{\frac{1}{2n}\sum_{i=1}^{n} X_i^2} \tag{12}$$

Using the law of large numbers we can say

$$\lim_{n\to\infty} \sqrt{\frac{1}{2n}\sum_{i=1}^{n} X_i^2} = \sqrt{\frac{1}{2n}\sum_{i=1}^{n} \mathbb{E}(X_i^2)} = \sqrt{\frac{1}{2n}2n\theta^2} = \theta \tag{13}$$

Using the 2nd moment of the Rayleigh distribution. Thus it is a consistent estimator.

1.3 Bias Check

The bias is defined as

$$\text{Bias}(\hat{\theta}) = \mathbb{E}(\hat{\theta}) - \theta \tag{14}$$

thus for our MLE estimator

$$\begin{aligned}
\text{Bias}(\hat{\theta}) &= \mathbb{E}\left(\sqrt{\frac{1}{2n}\sum_{i=1}^{n} X_i^2}\right) - \theta \qquad (15)\\
&= \frac{1}{\sqrt{2n}}\mathbb{E}\left(\sqrt{\sum_{i=1}^{n} X_i^2}\right) - \theta \qquad (16)
\end{aligned}$$

to find the expectation value we'll notice that if $X_i \sim \text{Rayleigh}(\theta)$ then $\sum X_i \sim \Gamma(n, 2\theta^2)$. We can use the formula for the moments of the gamma distribution for the 1/2-th moment

$$\mathbb{E}(\sqrt{\sum X_i}) = \sqrt{2}\theta\frac{\Gamma(n+\frac{1}{2})}{\Gamma(n)} \tag{17}$$

and inserting back we find

$$\begin{aligned}
\text{Bias}(\hat{\theta}) &= \frac{1}{\sqrt{2n}}\mathbb{E}\left(\sqrt{\sum_{i=1}^{n} X_i^2}\right) - \theta \qquad (18)\\
&= \frac{1}{\sqrt{2n}}\sqrt{2}\theta\frac{\Gamma(n+\frac{1}{2})}{\Gamma(n)} - \theta \qquad (19)\\
&= \theta\left(\frac{1}{\sqrt{n}}\frac{\Gamma(n+\frac{1}{2})}{\Gamma(n)} - 1\right) \qquad (20)
\end{aligned}$$

The expression in the brackets is non-zero so the estimator is biased.

1.4 Fisher Information and CRLB

The Fisher Information is defined by

$$
\begin{aligned}
I = -\mathbb{E}\left(\frac{\partial^2}{\partial\theta^2}\log f\right) &= -\mathbb{E}\left(\frac{\partial^2}{\partial\theta^2}\left(\log x - 2\log\theta - \frac{x^2}{2\theta^2}\right)\right) && (21)\\
&= -\mathbb{E}\left(\frac{\partial}{\partial\theta}\left(-\frac{2}{\theta} + \frac{x^2}{\theta^3}\right)\right) && (22)\\
&= -\mathbb{E}\left(\frac{2}{\theta^2} - \frac{3x^2}{\theta^4}\right) && (23)\\
&= -\left(\frac{2}{\theta^2} - \frac{3}{\theta^4}\mathbb{E}\left(x^2\right)\right) && (24)\\
&= -\left(\frac{2}{\theta^2} - \frac{3}{\theta^4}2\theta^2\right) && (25)\\
&= \frac{-2+6}{\theta^2} = \frac{4}{\theta^2} && (26)
\end{aligned}
$$

And since the Fisher information is based on expectation value it is linear and thus the fisher information of n i.i.d variables is

$$
I_n = \frac{4n}{\theta^2} \tag{27}
$$

The CRLB is therefore

$$
\mathrm{Var}(\hat{\theta}_{MLE}) \geq \frac{\theta^2}{4n} \tag{28}
$$

The MLE estimator is indeed asymptotically efficient as we saw in the lecture, but to show this we can use the asymptotic normalit property

$$
\sqrt{n}\left(\hat{\theta}_{MLE} - \theta\right) \to N(0, \frac{\theta^2}{4n}) \tag{29}
$$

therefore asymptotically

$$
\mathrm{Var}(\hat{\theta}_{MLE}) = \frac{\theta^2}{4n} \tag{30}
$$

which is indeed the CRLB.

2 Maximum A Posteriori Estimation with a Specific Prior

Assume the same Rayleigh distributed observations as in question 1, but now suppose tht prior knowledge about θ is modeled as an inverse-gamma distribution

$$p(\theta) = \frac{\beta^\alpha}{\Gamma(\alpha)} \theta^{-(\alpha+1)} e^{-\beta/\theta}, \quad \theta > 0$$

1. Derive the MAP esrtimator $\hat{\theta}_{MAP}$
2. Compare $\hat{\theta}_{ML}$ and $\hat{\theta}_{MAP}$ in terms of bias and asymptotic behavior.
3. Analyze how the choice of prior parameters α and β influences the MAP estimate.
4. Discuss the limiting cases $(\alpha, \beta \to 0)$ and $(\alpha, \beta \to \infty)$ and interpret their effects on the posterior estimate.

2.1 MAP Estimator

We are given a prior density, therefore the posterior is

$$\begin{aligned} p(\theta|x) &\propto f(x|\theta)p(\theta) && (31) \\ &= \left(\frac{x}{\theta^2} e^{-x^2/(2\theta^2)}\right)\left(\frac{\beta^\alpha}{\Gamma(\alpha)} \theta^{-(\alpha+1)} e^{-\beta/\theta}\right) && (32) \\ &= x\theta^{-(\alpha+3)} \frac{\beta^\alpha}{\Gamma(\alpha)} e^{-x^2/(2\theta^2)-\beta/\theta} && (33) \\ &\propto \theta^{-(\alpha+3)} \exp\left(-\frac{x^2}{2\theta^2} - \frac{\beta}{\theta}\right) && (34) \end{aligned}$$

The last line being correct because x and $\frac{\beta^\alpha}{\Gamma(\alpha)}$ are constants w.r.t θ. For ease, we'll work with the log-posterior, and introduce the sum to cover for the n i.i.d samples (noticing we need to multiply theta's power by n too)

$$l(\theta|x) = \log p(\theta|x) = -(\alpha + 2n + 1)\log\theta - \frac{\sum_{i=1}^{N} x_i^2}{2\theta^2} - \frac{\beta}{\theta} \tag{35}$$

therefore, marking $S = \sum x_i^2$

$$\begin{aligned} \frac{\partial l}{\partial \theta} = -\frac{\alpha + 2n + 1}{\theta} + \frac{S}{\theta^3} + \frac{\beta}{\theta^2} &\stackrel{!}{=} 0 && (36) \\ \to (\alpha + 2n + 1)\theta^2 - \beta\theta - S &\stackrel{!}{=} 0 && (37) \\ \to \theta_{1,2} = \frac{\beta \pm \sqrt{\beta^2 + 4(\alpha + 2n + 1) S}}{2\alpha + 4n + 2} & && (38) \end{aligned}$$

θ is positive therefore we only need the positive root. Therefore the MAP estimator is

$$\hat{\theta}_{\text{MAP}} = \frac{\beta + \sqrt{\beta^2 + 4(\alpha + 2n + 1)\sum x_i^2}}{2\alpha + 4n + 2} \tag{39}$$

2.2 Comparison to MLE & Prior Parameter Influence

Recalling the MLE is

$$\hat{\theta}_{MLE} = \sqrt{\frac{1}{2n}\sum_{i=1}^{n} X_i^2} \qquad\qquad \text{Bias}\left(\hat{\theta}_{MLE}\right) = \theta\left(\frac{1}{\sqrt{n}}\frac{\Gamma(n+\frac{1}{2})}{\Gamma(n)} - 1\right) \tag{40}$$

We can notice that while the MLE estimator is strictly biased to be smaller than the real value, the MAP estimator doesn't overtly have to be. The MAP estimator will heavily depend on the prior distribution's parameters α, β. If they are both quite small the MAP estimator will behave similarly to the MLE estimator, therefore will be biased to be below the real value, but with large values it will be biased to be above the real value.

$$\begin{aligned}
\lim_{n\to\infty} \hat{\theta}_{MAP} &= \lim_{n\to\infty} \frac{\beta + \sqrt{\beta^2 + 4\left(\alpha + 2n + 1\right)\sum x_i^2}}{2\alpha + 4n + 2} && (41)\\
&\approx \lim_{n\to\infty} \frac{\sqrt{\left(\alpha + 2n + 1\right)\sum x_i^2}}{2n} && (42)\\
&\approx \lim_{n\to\infty} \frac{\sqrt{2n\sum x_i^2}}{2n} && (43)\\
&= \lim_{n\to\infty} \sqrt{\frac{\sum x_i^2}{2n}} && (44)\\
&= \lim_{n\to\infty} \hat{\theta}_{MLE} && (45)
\end{aligned}$$

We already know the MLE estimator is consistent and since the MAP estimator at infinity behaves like the MLE estimator, it too much be consistent. For this reason both are also asymptotically unbiased, as the MLE estimator's bias tends to 0 at infinity. We can see that for the case $\alpha, \beta \to 0$ the MAP estimator effectively behaves just like the MLE estimator, which makes sense as that means the prior knowledge is effectively none. In the limit $\alpha, \beta \to \infty$ the effective meaning is that the prior distribution uniquely and completely determines the real value. It will mean the MAP estimator will effectively become a delta function at the real value with no variance. In the limit $n \to \infty$ we see the a-priori estimator becomes the same as the MLE estimator, which means the prior information we have becomes effectively irrelevant, since we can estimate the real value with arbitrary precision from the existing data.

3 Cramer-Rao Lower Bound and Efficiency

Let $(X_1, X_2, \ldots, X_n)$ be i.i.d samples from an exponential distribution

$$f(x, \lambda) = \lambda e^{-\lambda x}, \quad x \geq 0$$

where $\lambda > 0$ is unknown.

1. Derive the ML estimator $\hat{\lambda}_{ML}$.
2. Compute the Fisher Information.
3. Derive the CRLB for any unbiased estimator of λ.
4. Show that $\hat{\lambda}_{ML}$ achieves the CRLB and hence is an effective estimator.

1. Maximum Likelihood Estimator

Let $X_1, \ldots, X_n$ be i.i.d. samples from the exponential distribution

$$f(x; \lambda) = \lambda e^{-\lambda x}, \qquad x \geq 0, \ \lambda > 0.$$

The likelihood function is

$$L(\lambda) = \prod_{i=1}^{n} \lambda e^{-\lambda x_i} = \lambda^n e^{-\lambda \sum_{i=1}^{n} x_i}.$$

The log-likelihood is

$$\ell(\lambda) = n \log \lambda - \lambda \sum_{i=1}^{n} x_i.$$

Differentiating with respect to λ,

$$\frac{\partial \ell}{\partial \lambda} = \frac{n}{\lambda} - \sum_{i=1}^{n} x_i.$$

Setting the derivative equal to zero,

$$\frac{n}{\lambda} = \sum_{i=1}^{n} x_i \qquad \Longrightarrow \qquad \hat{\lambda}_{ML} = \frac{n}{\sum_{i=1}^{n} x_i} = \frac{1}{\bar{X}}.$$

Since the second derivative is

$$\frac{\partial^2 \ell}{\partial \lambda^2} = -\frac{n}{\lambda^2} < 0,$$

the solution is indeed a maximum. Hence

$$\boxed{\hat{\lambda}_{ML} = \frac{1}{\bar{X}}}.$$

2. Fisher Information

For one observation,

$$\log f(X;\lambda) = \log \lambda - \lambda X.$$

The first derivative is

$$\frac{\partial}{\partial \lambda} \log f(X;\lambda) = \frac{1}{\lambda} - X,$$

and the second derivative is

$$\frac{\partial^2}{\partial \lambda^2} \log f(X;\lambda) = -\frac{1}{\lambda^2}.$$

Thus the Fisher information for one sample is

$$I_1(\lambda) = -\mathbb{E}\left[\frac{\partial^2}{\partial \lambda^2} \log f(X;\lambda)\right] = \frac{1}{\lambda^2}.$$

For n independent samples,

$$I_n(\lambda) = nI_1(\lambda) = \frac{n}{\lambda^2}.$$

Therefore

$$\boxed{I(\lambda) = \frac{n}{\lambda^2}}.$$

3. Cramér–Rao Lower Bound

The Cramér–Rao inequality states that for any unbiased estimator $\hat{\lambda}$,

$$\mathrm{Var}(\hat{\lambda}) \geq \frac{1}{I_n(\lambda)} = \frac{1}{n/\lambda^2} = \frac{\lambda^2}{n}.$$

Hence the CRLB is

$$\boxed{\mathrm{Var}(\hat{\lambda}) \geq \frac{\lambda^2}{n}}.$$

4. Asymptotic Efficiency of the Maximum Likelihood Estimator

Let $S = \sum_{i=1}^{n} X_i$. Since the exponential distribution is memoryless, S has a Gamma(n, λ) distribution with density

$$f_S(s) = \frac{\lambda^n}{(n-1)!} s^{n-1} e^{-\lambda s}, \qquad s > 0.$$

The MLE can be written as

$$\hat{\lambda}_{ML} = \frac{n}{S}.$$

Bias computation

We compute

$$\mathbb{E}\left(\frac{1}{S}\right) = \frac{\lambda^n}{(n-1)!} \int_0^\infty s^{n-2} e^{-\lambda s}\, ds.$$

Using the Gamma integral

$$\int_0^\infty s^{k-1} e^{-\lambda s}\, ds = \frac{\Gamma(k)}{\lambda^k},$$

we obtain

$$\mathbb{E}\left(\frac{1}{S}\right) = \frac{\lambda}{n-1}.$$

Thus

$$\mathbb{E}(\hat{\lambda}_{ML}) = n\,\mathbb{E}\left(\frac{1}{S}\right) = \lambda \frac{n}{n-1}.$$

The estimator is slightly biased for finite n, but the bias vanishes as $n \to \infty$.

Variance computation

Similarly,

$$\mathbb{E}\left(\frac{1}{S^2}\right) = \frac{\lambda^n}{(n-1)!} \int_0^\infty s^{n-3} e^{-\lambda s}\, ds = \frac{\lambda^2}{(n-1)(n-2)}.$$

Therefore

$$\mathrm{Var}\left(\frac{1}{S}\right) = \frac{\lambda^2}{(n-1)(n-2)} - \left(\frac{\lambda}{n-1}\right)^2 = \frac{\lambda^2}{(n-2)(n-1)^2}.$$

Since $\hat{\lambda}_{ML} = n/S$,

$$\mathrm{Var}(\hat{\lambda}_{ML}) = n^2 \mathrm{Var}\left(\frac{1}{S}\right) = \frac{\lambda^2 n^2}{(n-2)(n-1)^2}.$$

As $n \to \infty$,

$$\mathrm{Var}(\hat{\lambda}_{ML}) \longrightarrow \frac{\lambda^2}{n}.$$

Conclusion

The CRLB is λ^2/n, and the variance of the MLE satisfies

$$\mathrm{Var}(\hat{\lambda}_{ML}) \sim \frac{\lambda^2}{n} \qquad \text{as } n \to \infty.$$

At the same time, the bias of the MLE tends to zero. Hence the MLE achieves the Cramér–Rao bound asymptotically and is therefore an asymptotically efficient estimator.

$$\boxed{\hat{\lambda}_{ML} \text{ is asymptotically efficient.}}$$

4 Neural Network Regression Implementation

The implementation for this is found in the included file **q4.ipynb**. This section will cover the discussion on comparison between the NN estimator and the linear regression.

For personal curiosity reasons we ran multiple NN architectures to see the effects of different depths and layer sizes. Of all the runs which can be found in detail in the jupyter notebook, the better performing ones all were similar in size and depth.

Method	MSE
Linear Regression	0.524
NN (8-32-32-1)	0.270
NN (8-32-16-1)	0.276
NN (8-16-16-16-1)	0.275

Table 1 – Different estimators for mean house value used on the scikit-learn California housing dataset and their mean square error. Neural Networks are designated NN and their architecture is described in parentheses with the first number being the input layer size and the final number the output layer size. The linear regression MSE was taken for the entire dataset while for the NN it was taken for the test dataset portion.

As can be seen from table **??**, the linear regression performed significantly worse than the neural networks. As we know from estimation theory, the MSE of an estimator can be written as

$$MSE = \text{Var} + \text{Bias}^2 \tag{46}$$

which for linear regression is more strongly affected by the bias, as it is not strongly affected by small changes in the data, since it can't approximately non-linear relationships in the data. For neural networks, the bias tends to be lower as they can fit data with more complex relationships but is more perceptible to variance increasing the MSE, such as in the case of overfitting.

Looking at the model NN (8-32-64-16-1) in the jupyter notebook we can see that it achieved a final training loss of 0.208 but its test loss was 0.281. This difference of 35% is quite significant and indicates possible overfitting. Additionally, model NN (8-262144-1) exhibited unstable behavior with its loss fluctuating wildly. In comparison, the best models' test loss was similar to their training and evaluation losses.